

VALIDATION REPORT:
PHOTON ATTENUATION SIMULATOR
(V1.04)

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1. EXECUTIVE SUMMARY

This report documents the validation of the Shielding Attenuation Simulator (V1.04). Version 1.04 adds Geometric Progression (G-P) gamma-ray buildup factor support for single-layer lead shielding cases. The previous validation report, V1.02, documented the Python rewrite of the original Java simulator and established correct operation of the narrow-beam Beer-Lambert attenuation model, NIST XCOM interpolation, inverse-square flux calculation, and robustness checks. That earlier report identified buildup factor integration as the next step for improving broad-beam shielding realism.

2. METHODOLOGY AND PHYSICS MODELS

2.1 Attenuation Model

The primary calculation for shielding effectiveness utilizes the Beer-Lambert Law. The simulator calculates the transmission (T) of a mono-energetic photon beam through N layers of material:

$$T = \exp\left(-\sum_{i=1}^N \mu_i x_i\right)$$

where:

- μ_i is the linear attenuation coefficient of the i-th material (cm^{-1}).
- x_i is the thickness of the i-th layer (cm).

2.2 Material Data and Interpolation

Attenuation coefficients are derived from NIST XCOM mass attenuation data (μ/ρ). For energies between tabulated data points, the simulator employs log-log interpolation to account for the power-law dependence of attenuation coefficients on energy.

$$\ln(\mu) = \ln(\mu_1) + \frac{\ln(E) - \ln(E_1)}{\ln(E_2) - \ln(E_1)} \cdot (\ln(\mu_2) - \ln(\mu_1))$$

2.3 Geometric Flux Calculation

The uncollided detector flux (ϕ) is calculated by combining the shield transmission with geometric spreading from a point source:

$$\phi_{\text{uncollided}} = \frac{S \cdot T}{4\pi r^2}$$

where:

- S is the source strength in photons / s

- r is the distance from the source to the detector (cm).
- T is the dimensionless shield transmission.

2.4 G-P Buildup Factor Model

Version 1.04 adds an optional buildup correction:

$$\phi_{buildup} = \phi_{uncollided} \cdot B$$

where:

- $\phi_{uncollided}$ is the narrow-beam uncollided flux
- B is the G-P buildup factor
- $\phi_{buildup}$ is the buildup-corrected flux estimate

The buildup factor is calculated using G-P fitting coefficients:

$$B = 1 + \frac{b-1}{K-1} (K^x - 1)$$

for:

$$K \neq 1$$

where:

- B is the buildup factor
- x is shield thickness in mean free paths
- b is a G-P fitting coefficient
- K is an intermediate G-P fitting value

The value of K is calculated as:

$$K = cx^a + d \cdot \frac{\tanh(\frac{x}{X_k}-2) - \tanh(-2)}{1 - \tanh(-2)}$$

where:

- x is the shield thickness in mean free paths
- c , a , X_k and d are G-P fitting coefficients
- $\tanh$ is the hyperbolic tangent function

When $K = 1$, the normal G-P buildup equation would involve division by zero. In this case, the simulator uses the limiting form:

$$B = 1 + (b - 1)x$$

for:

$$K = 1$$

where:

- B is the buildup factor
- x is shield thickness in mean free paths
- b is a G-P fitting coefficient
- K is an intermediate G-P fitting value

In the implementation, the simulator checks whether K is extremely close to 1.0 using a numerical tolerance. If:

$$|K - 1| < 10^{-12}$$

the limiting form is used instead of the standard G-P expression.

The implemented coefficient order is

material name, energy, b , c , a , X_0 , d

The current V1.04 implementation supports single-layer lead shielding only. This limitation is intentional because only lead G-P coefficients have been added and validated in this version.

2.5 Coefficient Source

The lead G-P coefficients used in V1.04 are from D. K. Trubey's ORNL/RSIC--49/R1; NUREG/CR-5740 report, *New Gamma-Ray Buildup Factor Data for Point Kernel Calculations: ANS-6.4.3 Standard Reference Data*. The OSTI record identifies the report as a 1991 Oak Ridge National Laboratory technical report and describes the ANS-6.4.3 evaluated buildup factor database for point-kernel calculations.

3. TEST CASE RESULTS

3.1 G-P Buildup Factor Calculation

The G-P coefficients for 0.6617 MeV were linearly interpolated between the 0.60 MeV and 0.80 MeV lead coefficient rows before evaluating the buildup factor.

Test Case	Input Parameters	Expected Result	Actual Result	Status
1	0.6617 MeV, 0 MFP	$B = 1.0$	$B = 1.0$	PASS
2	0.6617 MeV, 1 MFP	$B \approx b \approx 1.4576265$	$B \approx 1.4577876$	PASS

3	0.6617 MeV, 5 MFP	$B \approx 2.4215449$	$B \approx 2.4214919$	PASS
4	0.6617 MeV, 10 MFP	$B \approx 3.1637587$	$B \approx 3.1636799$	PASS

The zero-MFP test confirms that the buildup factor correctly returns 1.0 when no shielding thickness is present. The 1-MFP test confirms the G-P equation behavior where the buildup factor equals the b coefficient at one mean free path. The 5-MFP and 10-MFP tests confirm the buildup factor increases as shield thickness increases.

3.2 Buildup Corrected Flux Test

Test Case	Input Parameters	Expected Result	Actual Result	Status
5	5 cm, 0.6617 MeV, $S = 3.7 \times 10^{10}$ photons/s, $x = 100\text{cm}$	$\phi_{\text{buildup}} \approx$ 1.42612×10^3 photons/cm ² /s	$\phi_{\text{buildup}} \approx$ 1.426119×10^3 photons/cm ² /s	PASS

For this test case, the previously validated narrow-beam model gives:

$$T \approx 0.0018362592$$

$$\phi_{\text{uncollided}} \approx 5.40662 \times 10^2 \text{ photons/cm}^2/\text{s}$$

At 0.6617 MeV through 5 cm of lead:

$$x \approx 6.30002 \text{ MFP}$$

$$B \approx 2.63773$$

Therefore:

$$\phi_{\text{buildup}} = \phi_{\text{uncollided}} \cdot B$$

$$\phi_{\text{buildup}} \approx 1.426119 \times 10^3 \text{ photons/cm}^2/\text{s}$$

The result is physically consistent because the buildup corrected flux is greater than the uncollided narrow-beam flux.

3.3 Error Handling and Unsupported Case Tests

Test Case	Input Parameters	Expected Result	Actual Result	Status
6	Apply buildup to aluminum.	ValueError: G-P buildup mode currently supports	ValueError: G-P buildup mode currently supports	PASS

		only lead.	only lead.	
7	Apply buildup to 2 layers.	ValueError: G-P buildup mode currently supports only one shielding layer.	ValueError: G-P buildup mode currently supports only one shielding layer.	PASS
8	Request lead G-P coefficients at 0.01 MeV.	ValueError: Requested energy is outside the G-P coefficient data range.	ValueError: Requested energy is outside the G-P coefficient data range.	PASS
9	Pass -1 cm layer thickness.	ValueError: Layer thickness cannot be negative.	ValueError: Layer thickness cannot be negative.	PASS

4. ASSUMPTIONS AND LIMITATIONS

1. Single Material Lead Support Only: V1.04 supports G-P buildup correction only for single layer lead shielding. Other materials are not yet enabled.
2. No Multilayer Buildup Treatment: Multilayer buildup is not implemented. The simulator rejects buildup mode for multiple shielding layers because applying a single material buildup factor to mixed materials may produce misleading results.
3. Point-Source Geometry: The source is still modeled as an isotropic point source.
4. Narrow-Beam Transmission Still Used as Baseline: The buildup-corrected flux is calculated by multiplying the narrow-beam uncollided flux by the G-P buildup factor.
5. Coefficient Interpolation: G-P coefficients are linearly interpolated between tabulated energies. This is separate from the NIST XCOM attenuation interpolation, which uses log-log interpolation.
6. Exposure Buildup Coefficients: The implemented coefficients are exposure buildup coefficients. Future versions may distinguish exposure, dose, and energy buildup factors if additional coefficient tables are added.

5. CONCLUSION

The Shielding Attenuation Simulator V1.04 successfully implements and validates G-P buildup correction for single layer lead shielding. Validation tests confirmed coefficient interpolation, G-P buildup factor calculation, corrected flux calculation, and proper rejection of unsupported cases.

The V1.04 implementation represents a meaningful improvement over the V1.02 narrow-beam only model by adding a controlled first implementation of scattered photon buildup. The simulator remains intentionally limited to single layer lead buildup until additional material coefficient tables and multilayer buildup methods are implemented.

References

- [1] Y. Harima, Y. Sakamoto, S. Tanaka, and M. Kawai, “Validity of the Geometric-Progression Formula in Approximating Gamma-Ray Buildup Factors,” *Nuclear Science and Engineering*, vol. 94, no. 1, pp. 24–35, Sep. 1986. doi:10.13182/nse86-a17113
- [2] D. K. Trubey, “New gamma-ray buildup factor data for point kernel calculations: ANS-6.4.3 standard reference data,” IAEA, <https://inis.iaea.org/records/0arzw-ez976/files/20014493.pdf> (accessed Jul. 10, 2026).
- [3] Angstrom Sciences, “Density of elements chart – Angstrom Sciences elements density table,” Angstrom Sciences, <https://angstromsciences.com/density-elements-chart> (accessed Mar. 26, 2026).
- [4] J. Colledge, M. Wilczek, and C. Raymond, “Inverse square law | radiology reference article,” Radiopaedia, <https://radiopaedia.org/articles/inverse-square-law?lang=us> (accessed Mar. 26, 2026).
- [5] “Shielding Radiation Alphas, Betas, Gammas and Neutrons,” U.S.NRC, <https://www.nrc.gov/docs/ML1122/ML11229A721.pdf> (accessed Mar. 26, 2026).
- [6] M. J. Berger et al., “XCOM: Photon Cross Sections Database,” NIST, <https://dx.doi.org/10.18434/T48G6X> (accessed Mar. 26, 2026).